Decision Tree Learning: Complete Step-by-Step Notes

Example 6.3: Assess a student's performance during his course of study and predict whether a student will get a job offer or not in his final year of the course. The training dataset *T* consists of 10 data instances with attributes such as 'CGPA', 'Interactiveness', 'Practical Knowledge' and 'Communication Skills' as shown in Table 6.3. The target class attribute is the 'Job Offer'.

Table 6.3: Training Dataset T

S.No.	CGPA	Interactiveness	Practical Knowledge	Communication Skills	Job Offer
1.	≥ 9	Yes	Very good	Good	**Yes**
2.	≥ 8	No	Good	Moderate	**Yes**
3.	≥ 9	No	Average	Poor	**No**
4.	< 8	No	Average	Good	**No**
5.	≥ 8	Yes	Good	Moderate	**Yes**
6.	≥ 9	Yes	Good	Moderate	**Yes**
7.	< 8	Yes	Good	Poor	**No**
8.	≥ 9	No	Very good	Good	**Yes**
9.	≥ 8	Yes	Good	Good	**Yes**
10.	≥ 8	Yes	Average	Good	**Yes**

Solution

Step 1: Calculate the Entropy for the target class 'Job Offer'.

$$\text{Entropy_Info(Target Attribute = Job Offer)} = \text{Entropy_Info}(7, 3)$$
$$= -\left[(7/10) \log_2(7/10) + (3/10) \log_2(3/10) \right]$$
$$= -\left(-0.3599 + -0.5208 \right) = \mathbf{0.8807}$$

Iteration 1

Step 2: Calculate the Entropy_Info and Gain (Information Gain) for each of the attribute in the training dataset.

Table 6.4: Entropy Information for CGPA

CGPA	Job Offer = Yes	Job Offer = No	Total	Entropy
≥ 9	3	1	4	—
≥ 8	4	0	4	0
< 8	0	2	2	0

> Entropy_Info(T, CGPA) = (4/10)[-(3/4)$\log_2$(3/4) - (1/4)$\log_2$(1/4)] + (4/10)[-(4/4)$\log_2$(4/4) - 0] + (2/10)[0 - (2/2)$\log_2$(2/2)]
> = (4/10)(0.3111 + 0.4997) + 0 + 0 = **0.3243**
>
> Gain(CGPA) = 0.8807 - 0.3243 = **0.5564**

Table 6.5: Entropy Information for Interactiveness

Interactiveness	Job Offer = Yes	Job Offer = No	Total	Entropy
YES	5	1	6	—
NO	2	2	4	—

> Entropy_Info(T, Interactiveness) = (6/10)[-(5/6)$\log_2$(5/6) - (1/6)$\log_2$(1/6)] + (4/10)[-(2/4)$\log_2$(2/4) - (2/4)$\log_2$(2/4)]
> = (6/10)(0.2191 + 0.4306) + (4/10)(0.4997 + 0.4997) = 0.3898 + 0.3998 = **0.7896**
>
> Gain(Interactiveness) = 0.8807 - 0.7896 = **0.0911**

Table 6.6: Entropy Information for Practical Knowledge

Practical Knowledge	Job Offer = Yes	Job Offer = No	Total	Entropy
Very Good	2	0	2	0
Average	1	2	3	—
Good	4	1	5	—

> Entropy_Info(T, Practical Knowledge) = (2/10)[-(2/2)$\log_2$(2/2) - 0] + (3/10)[-(1/3)$\log_2$(1/3) - (2/3)$\log_2$(2/3)] + (5/10)[-(4/5)$\log_2$(4/5) - (1/5)$\log_2$(1/5)]
> = (2/10)(0) + (3/10)(0.5280 + 0.3897) + (5/10)(0.2574 + 0.4641) = 0 + 0.2753 + 0.3608 = **0.6361**
>
> Gain(Practical Knowledge) = 0.8807 - 0.6361 = **0.2446**

Table 6.7: Entropy Information for Communication Skills

Communication Skills	Job Offer = Yes	Job Offer = No	Total	Entropy
Good	4	1	5	—
Moderate	3	0	3	0
Poor	0	2	2	0

$$\text{Entropy_Info}(T, \text{Communication Skills}) = (5/10)[\ -(4/5)\log_2(4/5) - (1/5)\log_2(1/5)\ ] + (3/10)[\ -(3/3)\log_2(3/3) - 0\ ] + (2/10)[\ 0 - (2/2)\log_2(2/2)\ ]$$
$$= (5/10)(0.2574 + 0.4641) + (3/10)(0) + (2/10)(0) = \mathbf{0.3609}$$

$$\text{Gain}(\text{Communication Skills}) = 0.8813 - 0.36096 = \mathbf{0.5203}$$

Table 6.8: Gain

Attributes	Gain
CGPA	**0.5564**
Interactiveness	0.0911
Practical Knowledge	0.2246
Communication Skills	0.5203

Step 3: From Table 6.8, choose the attribute for which entropy is minimum and therefore the Gain is maximum as the best split attribute. The best split attribute is **CGPA** since it has the maximum gain. So, we choose CGPA as the root node. There are three distinct values for CGPA with outcomes ≥ 9, ≥ 8 and < 8. The entropy value is 0 for ≥ 8 and < 8 with all instances classified as Job Offer = Yes for ≥ 8 and Job Offer = No for < 8. Hence, both ≥ 8 and < 8 end up in a leaf node. The tree grows with the subset of instances CGPA ≥ 9 as shown in Figure 6.3.

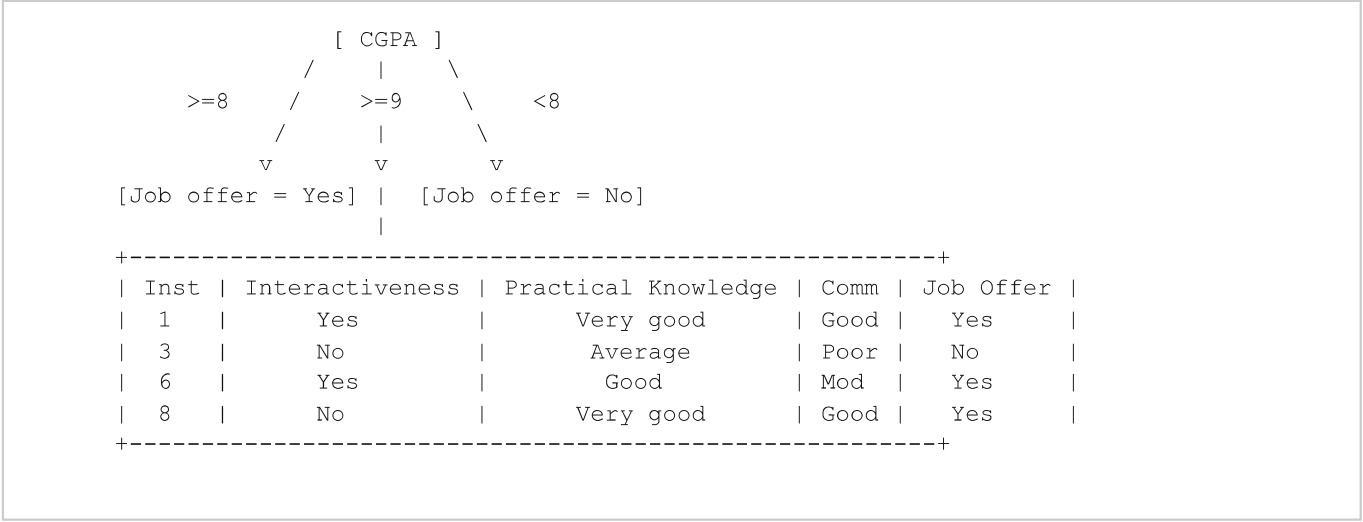


Figure 6.3: Decision Tree After Iteration 1

Iteration 2

Now, continue the same process for the subset of data instances branched with CGPA ≥ 9. In this iteration, the same process of computing the Entropy_Info and Gain are repeated with the subset of training set. The subset consists of 4 data instances as shown in the above Figure 6.3.

$$Entropy_Info(T_{sub}) = Entropy_Info(3, 1)$$
$$= - [(3/4) \log_2(3/4) + (1/4) \log_2(1/4)]$$
$$= - (-0.3111 + -0.4997) = \mathbf{0.8108}$$

$$Entropy_Info(T, Interactiveness) = (2/4)[-(2/2)\log_2(2/2) - 0] + (2/4)[-(1/2)\log_2(1/2) - (1/2)\log_2(1/2)]$$
$$= 0 + 0.4997 = \mathbf{0.4997}$$
$$Gain(Interactiveness) = 0.8108 - 0.4997 = \mathbf{0.3111}$$

$$Entropy_Info(T, Practical\ Knowledge) = (2/4)[-(2/2)\log_2(2/2) - 0] + (1/4)[0 -(1/1)\log_2(1/1)] + (1/4)[-(1/1)\log_2(1/1) - 0] = \mathbf{0}$$
$$Gain(Practical\ Knowledge) = 0.8108 - 0 = \mathbf{0.8108}$$

$$Entropy_Info(T, Communication\ Skills) = (2/4)[-(2/2)\log_2(2/2) - 0] + (1/4)[0 -(1/1)\log_2(1/1)] + (1/4)[-(1/1)\log_2(1/1) - 0] = \mathbf{0}$$
$$Gain(Communication\ Skills) = 0.8108 - 0 = \mathbf{0.8108}$$

Table 6.9: Total Gain

Attributes	Gain
Interactiveness	0.3111
Practical Knowledge	**0.8108**
Communication Skills	**0.8108**

Here, both the attributes 'Practical Knowledge' and 'Communication Skills' have the same Gain. So, we can either construct the decision tree using 'Practical Knowledge' or 'Communication Skills'. The final decision tree is shown in Figure 6.4.

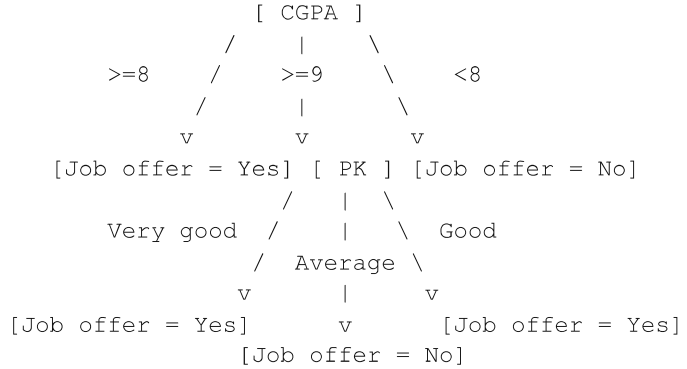


Figure 6.4: Final Decision Tree

Extracted Decision Rules

- **Rule 1:** IF (CGPA ≥ 8) THEN **Job offer = Yes**
- **Rule 2:** IF (CGPA < 8) THEN **Job offer = No**
- **Rule 3:** IF (CGPA ≥ 9) AND (Practical Knowledge = Average) THEN **Job offer = No**
- **Rule 4:** IF (CGPA ≥ 9) AND (Practical Knowledge ∈ {Good, Very good}) THEN **Job offer = Yes**